

R Notebook

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0      v readr     2.1.5
## v ggplot2    4.0.0      v stringr  1.5.1
## v lubridate  1.9.4      v tibble   3.3.0
## v purrr      1.0.4      v tidyr    1.3.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(purrr)
library(ggplot2)
library(vegan)
```

```
## Loading required package: permute
```

```
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##   expand, pack, unpack
```

```
library(stringr)
library(patchwork)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##   group_rows
```

```
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
knitr::opts_chunk$set(dev = "cairo_pdf")
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
```

Sample Taxon Reports

```
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_a",
                        full.names = TRUE)
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m")

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# separate slurry from pond samples
pond_samples = c("s9", "s10", "s19", "s20")

# Create sample metadata using helper function
sample_metadata = create_sample_metadata(seq_metrics, pond_samples) %>%
  select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species <- merged_data %>%
  filter(tax_level == 1) %>%
  filter(experiment == 3) %>% # filter in specific experiments
  filter(water_control == "No") %>%
  filter(infection_class != "Definite") # filter out positive controls

species_data_filtered = all_data_species %>%
  filter(nt_rpm >= 1 | nr_rpm >= 1 | nt_count >= 3) %>% # positive inclusion criteria
  filter(nt_alignment_length > 50)
```

```

# specific names that should group together
ent_v = "Enterovirus G"
circo_v = "Circovirus sp."
sapo_v = paste(c("Sapovirus", "Sapporo"), collapse = "|")
tes_v = "Teschovirus"
aich_v = "Aichivirus"
astr_v = paste(c("Porcine astrovirus", "Mamastrovirus 3", "Mamastrovirus 22", "Mamastrovirus pig", "PoAstV", "Pig astrovirus CX1"), collapse = "|")
hvp18 = "Alphapapillomavirus"
sapel_v = paste(c("Sapelovirus sp.", "Sapelovirus A"), collapse = "|")
boc_v = paste(c("Porcine bocavirus", "Bocaparvovirus", "Bocavirus pig"), collapse = "|")

# Virus pattern definitions
patterns <- c(
  EVG      = ent_v,
  PCV      = circo_v,
  PSapoV   = sapo_v,
  PTeV     = tes_v,
  PKV      = aich_v,
  PAstV    = astr_v,
  #HPV18   = hvp18,
  PSapeV   = sapel_v,
  PBoV     = boc_v
)

```

```

# Extract sample metadata (one row per sample)
sample_meta <- all_data_species %>%
  select(sample_name, library_prep, experiment, sample_type) %>%
  distinct()

# Create wide count matrix: taxa x samples
#count_mat <- species_data_filtered %>%
count_mat = all_data_species %>%
  group_by(sample_name, name) %>%
  summarise(reads = mean(nt_count, na.rm = TRUE), .groups = "drop") %>% # counting reads for each taxa
  pivot_wider(names_from = sample_name, values_from = reads, values_fill = 0) %>%
  column_to_rownames("name")

```

```

# ---- USER PARAMETERS ----
#depths <- c(5e5, 1e6, 2e6, 3e6, 4e6, 5e6, 7e6, 9e6, 11e6) # Target depths to simulate
depths = c(1e5, 2e5, 3e5, 4e5, 5e5, 6e5, 7e5, 8e5, 9e5, 1e6, 1.1e6, 1.2e6, 1.3e6, 1.4e6, 1.5e6)
n_reps <- 1 # Replicates per depth

# Start the simulation!
res <- simulate_downsampling(count_mat, depths, n_reps)

```

```

# ---- SUMMARIZE RESULTS ----
# Diversity summary
div_summary <- res %>%
  group_by(depth) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),

```

```

    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# By library prep
div_summary_libprep <- res %>%
  group_by(depth, library_prep) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),
    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# By sample type
div_summary_sampletype <- res %>%
  group_by(depth, sample_type) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),
    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# remove _M from samples names for comparison
res = res %>%
  mutate(sample = str_remove(sample, "_M.*$"))

res_agg <- res %>%
  rename(richness_val = richness) %>%
  group_by(sample, library_prep, depth) %>%
  summarise(
    richness_mean = mean(richness_val),
    shannon_mean = mean(shannon),
    .groups = "drop"
  )

# do friedman test measuring richness by library prep pairing by sample
friedman_results <- res_agg %>%
  group_by(depth) %>%
  summarise(
    test = list(
      friedman.test(richness_mean ~ library_prep | sample)
    ),
    .groups = "drop"
  ) %>%
  mutate(
    statistic = map_dbl(test, ~ .$statistic),
    p_value = map_dbl(test, ~ .$p.value)
  )

```

```

# Join friedman results back to the overall table
div_summary = div_summary %>%
  left_join(friedman_results, by = "depth")

div_summary = div_summary %>%
  select(-shannon_sd, -richness_sd, -test, -statistic)

div_summary

```

```

## # A tibble: 15 x 4
##   depth richness_mean shannon_mean p_value
##   <dbl>         <dbl>         <dbl>   <dbl>
## 1 100000         4852.           5.87 0.00157
## 2 200000         6262.           5.89 0.00157
## 3 300000         7163.           5.90 0.00157
## 4 400000         7739.           5.90 0.00157
## 5 500000         8169.           5.90 0.0114
## 6 600000         8516.           5.90 0.0114
## 7 700000         8776.           5.91 0.0578
## 8 800000         8970.           5.91 0.0578
## 9 900000         9119.           5.91 0.0578
##10 1000000        9261.           5.91 0.0578
##11 1100000        9376.           5.91 0.206
##12 1200000        9471.           5.91 0.206
##13 1300000        9558.           5.91 0.206
##14 1400000        9637.           5.91 0.206
##15 1500000        9704.           5.91 0.206

```

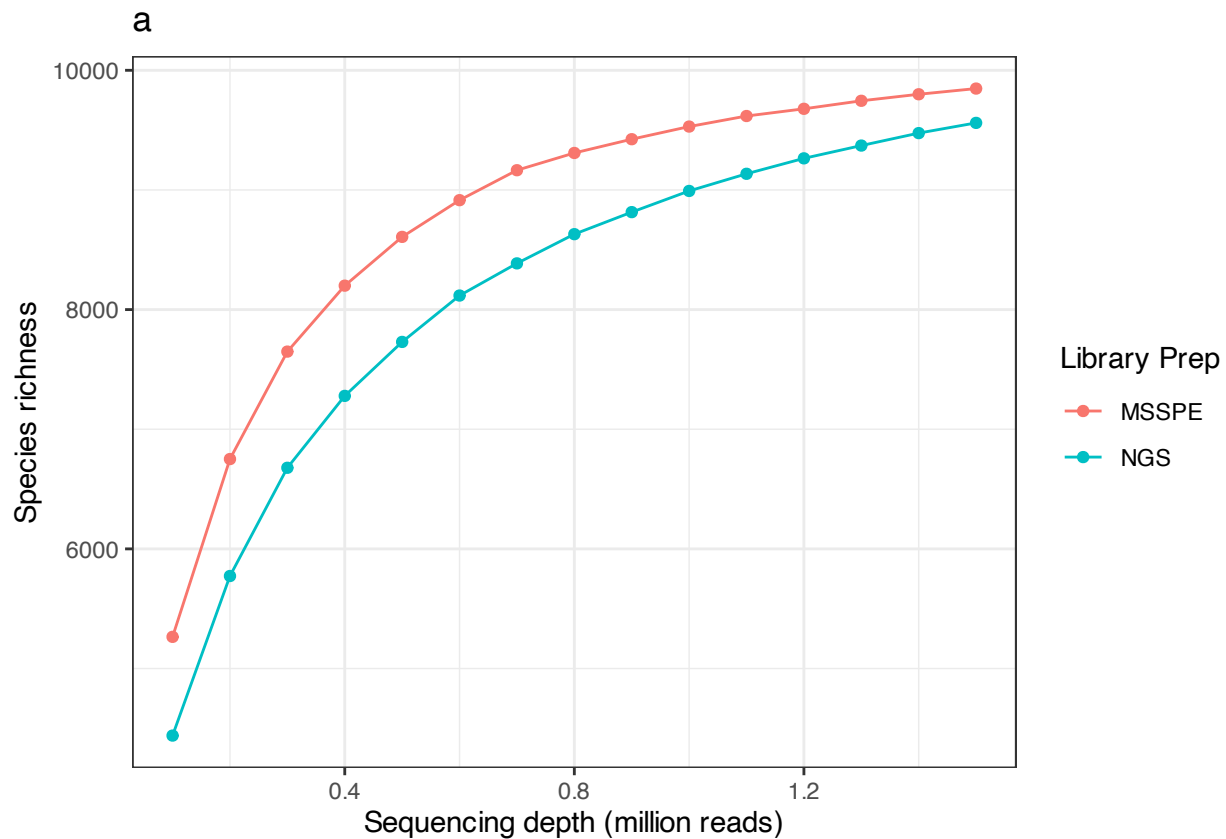
Rarefaction curves

```

div_summary_libprep_filter = div_summary_libprep %>%
  filter(depth <= 7.0e+06) %>%
  rename("Library Prep" = library_prep)

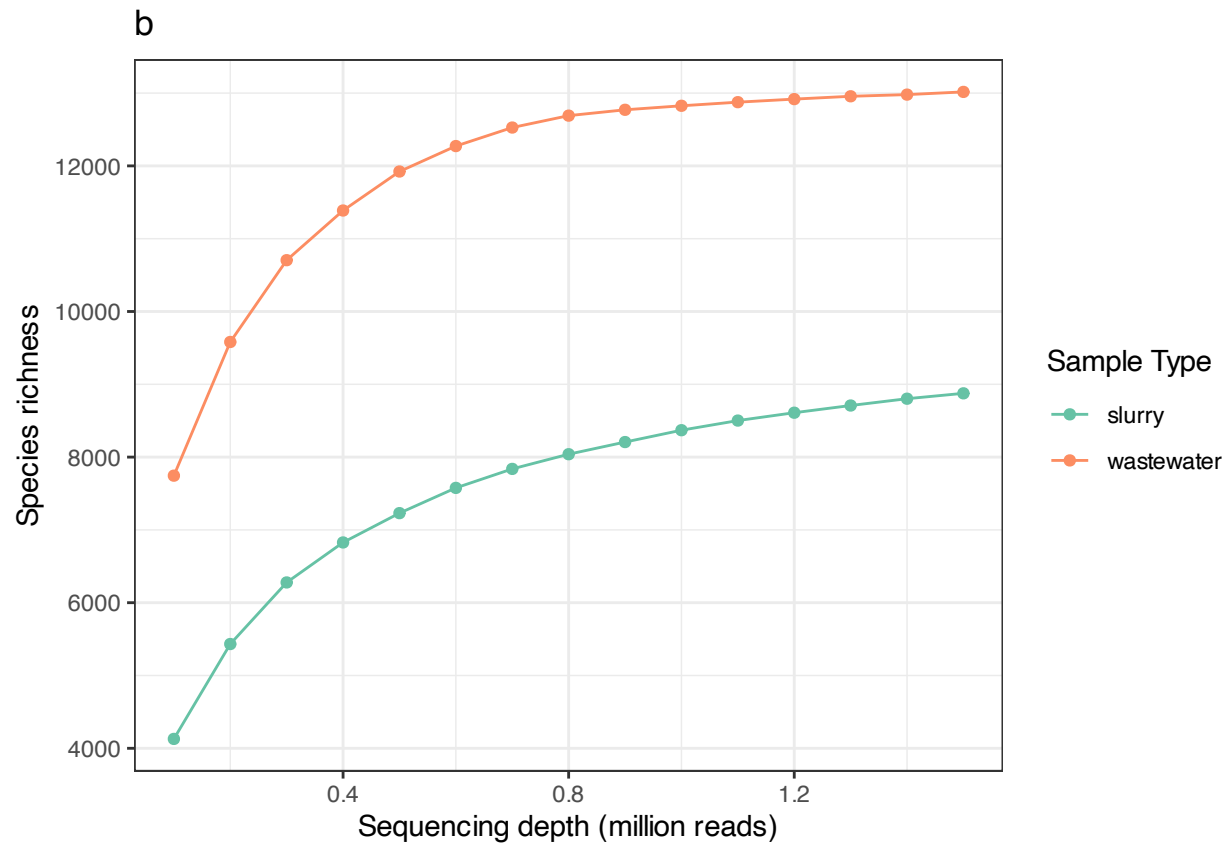
# Plot Rarefaction Curve
p1 <- ggplot(div_summary_libprep_filter, aes(x = depth/1e6, y = richness_mean, fill = `Library Prep`, c
  geom_line(aes(y = richness_mean)) +
  geom_point(aes(y = richness_mean)) +
  #geom_ribbon(aes(ymin = richness_mean - richness_sd, ymax = richness_mean + richness_sd), alpha = 0.2)
  labs(x = "Sequencing depth (million reads)", y = "Species richness",
    title = "a") +
  theme_bw()
p1

```

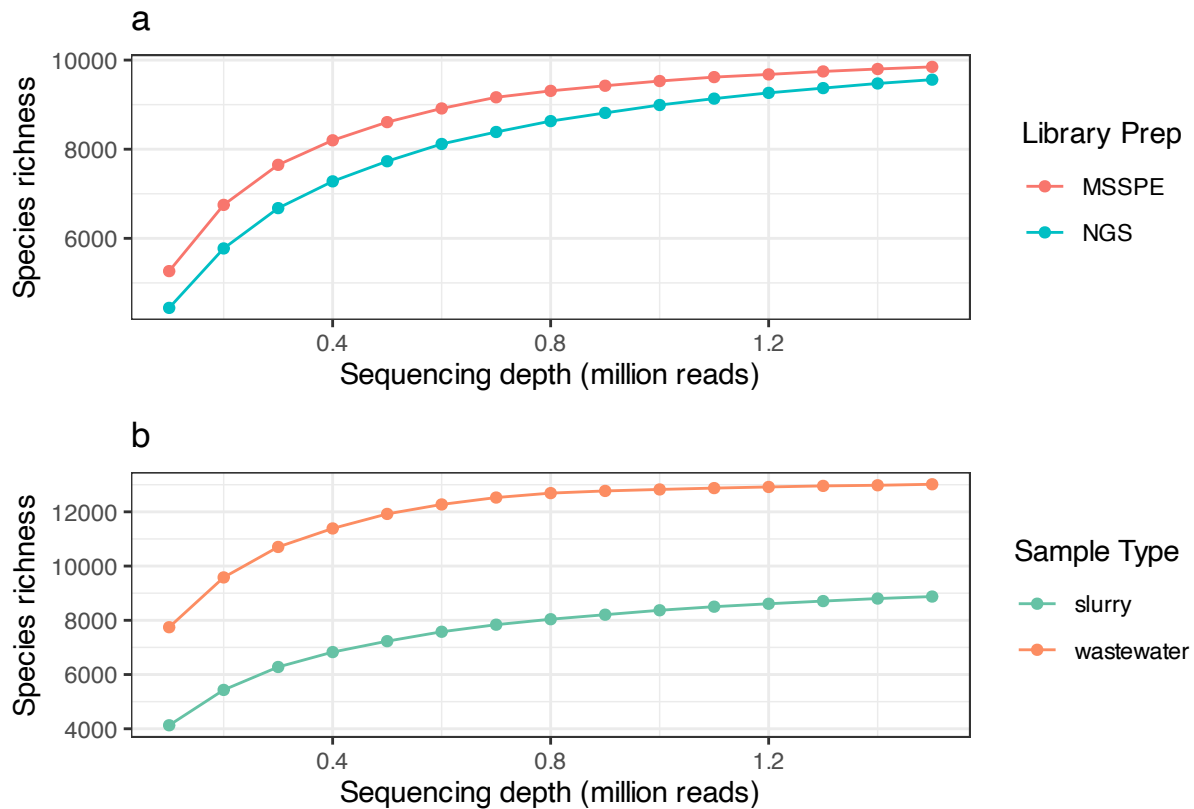


```
div_summary_sampletype_filter = div_summary_sampletype %>%
  filter(depth <= 7.0e+06) %>%
  mutate(
    sample_type = recode(sample_type, `environmental` = "wastewater")
  ) %>%
  rename("Sample Type" = sample_type)

# Plot Rarefaction Curve
p2 <- ggplot(div_summary_sampletype_filter, aes(x = depth/1e6, y = richness_mean, color = `Sample Type`)) +
  geom_line(aes(y = richness_mean)) +
  geom_point(aes(y = richness_mean)) +
  scale_color_brewer(palette = "Set2") +
  #geom_ribbon(aes(ymin = richness_mean - richness_sd, ymax = richness_mean + richness_sd), alpha = 0.2)
  labs(x = "Sequencing depth (million reads)", y = "Species richness",
    title = "b") +
  theme_bw()
p2
```



```
combined_curves = p1 / p2  
combined_curves
```



TARGET-SPECIFIC SEQ DEPTH ANALYSIS

```
# Start the simulation!
suppressWarnings(
  res_targets <- simulate_downsampling_targets(
    count_mat,
    depths = c(1e5, 2e5, 3e5, 4e5, 5e5, 6e5, 7e5, 8e5, 9e5, 1e6, 1.1e6, 1.2e6, 1.3e6, 1.4e6, 1.5e6),
    n_reps = 10,
    patterns
  )
)

# Function to calculate detection probabilities from rarefied reads
summarize_detection <- function(res_targets, detection_threshold) {

  # 1. Pivot longer: each row = sample x replicate x target
  target_cols <- names(res_targets)[
    !(names(res_targets) %in% c("sample", "depth", "rep", "library_prep", "experiment", "sample_type"))
  ]

  det_long <- res_targets %>%
    pivot_longer(
      cols = all_of(target_cols),
      names_to = "target",
      values_to = "total_reads"
    ) %>%
    mutate(
      detected = as.integer(total_reads >= detection_threshold)
    )
}
```



```

)

# 2. Detection probability per depth and target
det_summary <- det_long %>%
  group_by(depth, target) %>%
  summarise(
    mean_total_reads = mean(total_reads, na.rm = TRUE),
    detect_prob = mean(detected, na.rm = TRUE),
    n_reps = n(),
    .groups = "drop"
  )

# 3. Detection probability by library prep
det_summary_libprep <- det_long %>%
  group_by(depth, library_prep, target) %>%
  summarise(
    mean_total_reads = mean(total_reads, na.rm = TRUE),
    detect_prob = mean(detected, na.rm = TRUE),
    n_reps = n(),
    .groups = "drop"
  )

list(
  long = det_long,
  summary = det_summary,
  summary_libprep = det_summary_libprep
)
}

# Summarized detection probabilities from simulation
res_summary <- summarize_detection(res_targets, detection_threshold = 5)

# Access outputs
det_long <- res_summary$long # replicate-level total reads + detected 0/1
det_summary <- res_summary$summary # mean reads + detection prob per depth x target
det_summary_libprep <- res_summary$summary_libprep # mean reads + detection prob per depth x target x

```

Linear Mixed Model for Diversity

```
library(glmmTMB)
```

```
## Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
## glmmTMB was built with TMB package version 1.9.17
## Current TMB package version is 1.9.18
## Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more)

```

```
library(performance)
```

```

# get response data from simulation
model_div_data = res %>%
  mutate(
    depth_scaled = depth / 1e5, # scale depth

```

```

library_prep = factor(library_prep), # convert to factor
library_prep = relevel(library_prep, ref = "NGS") # set NGS as reference
)

# create base model for diversity metrics
model_seq_depth_div <- glmmTMB(
  richness ~ library_prep * depth_scaled * sample_type + (1 | sample),
  data = model_div_data,
  family = nbinom2() # use if Richness
)

# print summary of model performance
summary(model_seq_depth_div)

```

```

## Family: nbinom2 ( log )
## Formula:
## richness ~ library_prep * depth_scaled * sample_type + (1 | sample)
## Data: model_div_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    5024.6    5061.6   -2502.3    5004.6      290
##
## Random effects:
##
## Conditional model:
##   Groups Name      Variance Std.Dev.
## sample (Intercept) 0.04553  0.2134
## Number of obs: 300, groups: sample, 10
##
## Dispersion parameter for nbinom2 family (): 72.4
##
## Conditional model:
##                                     Estimate Std. Error z value
## (Intercept)                       9.085500   0.157913   57.53
## library_prepMSSPE                  0.180287   0.065733    2.74
## depth_scaled                      0.031533   0.005158    6.11
## sample_typeslurry                 -0.611263   0.176575   -3.46
## library_prepMSSPE:depth_scaled    -0.011765   0.007277   -1.62
## library_prepMSSPE:sample_typeslurry -0.035849   0.073611   -0.49
## depth_scaled:sample_typeslurry     0.014853   0.005777    2.57
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.004085   0.008152    0.50
##                                     Pr(>|z|)
## (Intercept)                       < 2e-16 ***
## library_prepMSSPE                  0.006093 **
## depth_scaled                      9.76e-10 ***
## sample_typeslurry                 0.000537 ***
## library_prepMSSPE:depth_scaled     0.105959
## library_prepMSSPE:sample_typeslurry 0.626255
## depth_scaled:sample_typeslurry     0.010138 *
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.616323
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
# or profile CI (more accurate but slower)
exp(confint(model_seq_depth_div, parm="beta_", method="profile"))
```

```
##                                2.5 %      97.5 %
## (Intercept)                   6297.6586695 1.237088e+04
## library_prepMSSPE              1.0522229 1.362830e+00
## depth_scaled                   1.0216137 1.042561e+00
## sample_typeslurry              0.3720260 7.914728e-01
## library_prepMSSPE:depth_scaled  0.9742489 1.002550e+00
## library_prepMSSPE:sample_typeslurry 0.8346620 1.115074e+00
## depth_scaled:sample_typeslurry  1.0034878 1.026560e+00
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.9881121 1.020319e+00
```

```
# overdispersion
overdisp_fun(model_seq_depth_div)
```

```
##      chisq      ratio      rdf      p
## 265.8864401  0.9168498 290.0000000 0.8419631
```

Microbial diversity sequencing depth tables

```
cat("\\captionsetup{labelformat=empty}")
```

```
# Fixed effects table
fixed_tbl_div <- summary(model_seq_depth_div)$coefficients$cond %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  rename(Estimate = Estimate, SE = `Std. Error`, z = `z value`, p = `Pr(>|z|)`)

# Add odds ratio and 95% CI
fixed_tbl_div <- fixed_tbl_div %>%
  mutate(
    OR = exp(Estimate),
    CI_lower = exp(Estimate - 1.96*SE),
    CI_upper = exp(Estimate + 1.96*SE),
    Percent_change = (OR - 1) * 100,
    Effect = recode(Effect,
      `(Intercept)` = "intercept",
      `library_prepMSSPE` = "Library prep",
      `depth_scaled` = "Sequencing depth",
      `sample_typeslurry` = "Sample type",
      `library_prepMSSPE:depth_scaled` = "Lib prep * Sequencing depth",
      `library_prepMSSPE:sample_typeslurry` = "Lib prep * Sample type",
      `depth_scaled:sample_typeslurry` = "Sequencing depth * Sample type",
      `library_prepMSSPE:depth_scaled:sample_typeslurry` = "Lib Prep * Depth * Sample Type"
    )
  ) %>%
  rename(
    "Percent change" = Percent_change,
    "CI 25%" = CI_lower,
    "CI 75%" = CI_upper
  )
```

```

# Random effects table
# Extract variance components
vc <- VarCorr(model_seq_depth_div)

random_tbl_div <- data.frame(
  Random_Effect = c("Virus target", "Sample"),
  Variance = c(vc$cond$target[1,1], vc$cond$sample[1,1]),
  SD = sqrt(c(vc$cond$target[1,1], vc$cond$sample[1,1]))
)

# Combine in a nice list for reporting
list(Fixed_Effects = fixed_tbl_div, Random_Effects = random_tbl_div)

```

```

$Fixed_Effects Effect Estimate SE z 1 intercept 9.085499958 0.157912786 57.5349228 2 Library prep
0.180287221 0.065732766 2.7427299 3 Sequencing depth 0.031533457 0.005158129 6.1133520 4 Sample type
-0.611263257 0.176575404 -3.4617690 5 Lib prep * Sequencing depth -0.011764980 0.007277499 -1.6166241 6
Lib prep * Sample type -0.035849136 0.073611477 -0.4870047 7 Sequencing depth * Sample type 0.014853324
0.005777054 2.5710897 8 Lib Prep * Depth * Sample Type 0.004084552 0.008151681 0.5010686 p OR CI
25% CI 75% Percent change 1 0.000000e+00 8826.3774762 6476.8586579 1.202820e+04 8.825377e+05
2 6.093079e-03 1.1975613 1.0527976 1.362231e+00 1.975613e+01 3 9.755982e-10 1.0320359 1.0216547
1.042523e+00 3.203590e+00 4 5.366376e-04 0.5426649 0.3839085 7.670713e-01 -4.573351e+01 5 1.059594e-
01 0.9883040 0.9743070 1.002502e+00 -1.169604e+00 6 6.262550e-01 0.9647858 0.8351636 1.114526e+00
-3.521417e+00 7 1.013791e-02 1.0149642 1.0035365 1.026522e+00 1.496418e+00 8 6.163228e-01 1.0040929
0.9881777 1.020264e+00 4.092905e-01

```

```

$Random_Effects Random_Effect Variance SD 1 Virus target 0.04553455 0.2133883 2 Sample 0.04553455
0.2133883

```

```

# Optional: use kable() to print nicely in RMarkdown
kable(fixed_tbl_div, digits = 3, caption = "Fixed effects from GLMM (negative binomial)")

```

Fixed effects from GLMM (negative binomial)

Effect	Estimate	SE	z	p	OR	CI 25%	CI 75%	Percent change
intercept	9.085	0.158	57.535	0.000	8826.377	6476.859	12028.198	882537.748
Library prep	0.180	0.066	2.743	0.006	1.198	1.053	1.362	19.756
Sequencing depth	0.032	0.005	6.113	0.000	1.032	1.022	1.043	3.204
Sample type	-0.611	0.177	-3.462	0.001	0.543	0.384	0.767	-45.734
Lib prep * Sequencing depth	-0.012	0.007	-1.617	0.106	0.988	0.974	1.003	-1.170
Lib prep * Sample type	-0.036	0.074	-0.487	0.626	0.965	0.835	1.115	-3.521
Sequencing depth * Sample type	0.015	0.006	2.571	0.010	1.015	1.004	1.027	1.496
Lib Prep * Depth * Sample Type	0.004	0.008	0.501	0.616	1.004	0.988	1.020	0.409

```

kable(random_tbl_div, digits = 3, caption = "Random effects (variance components) from GLMM")

```

Random effects (variance components) from GLMM

Random_Effect	Variance	SD
Virus target	0.046	0.213
Sample	0.046	0.213

Linear Mixed Model for Targets

```
library(glmmTMB)
library(broom.mixed)
library(performance)

det_long <- det_long %>%
  mutate(
    depth_scaled = depth / 1e5,          # scale depth
    library_prep = factor(library_prep), # convert to factor
    library_prep = relevel(library_prep, ref = "NGS") # set NGS as reference
  )

# create base model for detection probability
model_seq_depth <- glmmTMB(
  detected ~ library_prep * depth_scaled + (1 | target) + (1 | sample),
  data = det_long,
  family = binomial(link = "logit")
)

# print summary of model performance
summary(model_seq_depth)
```

```
## Family: binomial ( logit )
## Formula:      detected ~ library_prep * depth_scaled + (1 | target) + (1 |
##      sample)
## Data: det_long
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##   9965.0   10013.5   -4976.5    9953.0     23994
##
## Random effects:
##
## Conditional model:
## Groups Name      Variance Std.Dev.
## target (Intercept) 22.106   4.702
## sample (Intercept)  7.605   2.758
## Number of obs: 24000, groups: target, 8; sample, 20
##
## Conditional model:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.925166   1.902222   0.486    0.627
## library_prepMSSPE 1.972029   1.238717   1.592    0.111
## depth_scaled      0.256004   0.009238  27.713 < 2e-16 ***
## library_prepMSSPE:depth_scaled -0.081015   0.012557  -6.452 1.11e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# or profile CI (more accurate but slower)
exp(confint(model_seq_depth, parm="beta_", method="profile"))
```

```
##                2.5 %      97.5 %
## (Intercept)      0.04541439 191.6992340
## library_prepMSSPE 0.56050052  92.6568814
## depth_scaled     1.26873234   1.3155378
## library_prepMSSPE:depth_scaled 0.89971271   0.9451304
```

```
# standard error
summary(model_seq_depth)$coefficients$cond
```

```
##                Estimate Std. Error   z value    Pr(>|z|)
## (Intercept)      0.9251658 1.90222190  0.4863606 6.267115e-01
## library_prepMSSPE 1.9720295 1.23871733  1.5919931 1.113863e-01
## depth_scaled     0.2560039 0.00923779 27.7126745 4.912278e-169
## library_prepMSSPE:depth_scaled -0.0810149 0.01255696 -6.4517919 1.105353e-10
## attr(,"ddf")
## [1] "asymptotic"
```

```
# check residuals
r2_model <- r2_nakagawa(model_seq_depth)
r2_model
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.904
##   Marginal R2: 0.039
```

```
# check overdispersion
overdisp_fun(model_seq_depth)
```

```
##      chisq      ratio      rdf      p
## 37287.983506    1.554054 23994.000000 0.000000
```

```
# Fit zero-inflated model for comparison
model_zi <- glmmTMB(
  detected ~ library_prep * depth_scaled + (1|target) + (1|sample),
  ziformula = ~1,
  data = det_long,
  family = binomial
)
```

```
# Compare AIC
AIC(model_seq_depth, model_zi)
```

```
##      df      AIC
## model_seq_depth 6 9964.959
## model_zi        7 9927.335
```

```

# Compare alternative sequencing depth modeling

# try log transformed depth
det_long <- det_long %>%
  mutate(log_depth = log(depth))

model_log <- glmmTMB(
  detected ~ library_prep * log_depth + (1|target) + (1|sample),
  data = det_long,
  family = binomial
)
# print AIC comparison
AIC(model_seq_depth, model_log)

```

```

##              df      AIC
## model_seq_depth 6 9964.959
## model_log        6 9446.980

```

Viral target sequencing depth tables

```

cat("\\captionsetup{labelformat=empty}")

```

```

# Fixed effects table
fixed_tbl <- summary(model_seq_depth)$coefficients$cond %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  rename(Estimate = Estimate, SE = `Std. Error`, z = `z value`, p = `Pr(>|z|)`)

# Add odds ratio and 95% CI
fixed_tbl <- fixed_tbl %>%
  mutate(
    OR = exp(Estimate),
    CI_lower = exp(Estimate - 1.96*SE),
    CI_upper = exp(Estimate + 1.96*SE),
    Effect = recode(Effect,
      `(Intercept)` = "intercept",
      `library_prepMSSPE` = "Library prep",
      `depth_scaled` = "Sequencing depth",
      `library_prepMSSPE:depth_scaled` = "Library prep * Sequencing depth"
    )
  ) %>%
  rename(
    "CI 25%" = CI_lower,
    "CI 75%" = CI_upper
  )

# Random effects table
# Extract variance components
vc <- VarCorr(model_seq_depth)

random_tbl <- data.frame(
  Random_Effect = c("Virus target", "Sample"),

```

```

Variance = c(vc$cond$target[1,1], vc$cond$sample[1,1]),
SD = sqrt(c(vc$cond$target[1,1], vc$cond$sample[1,1]))
)

# Combine in a nice list for reporting
list(Fixed_Effects = fixed_tbl, Random_Effects = random_tbl)

```

```

$Fixed_Effects Effect Estimate SE z 1 intercept 0.9251658 1.90222190 0.4863606 2 Library prep 1.9720295
1.23871733 1.5919931 3 Sequencing depth 0.2560039 0.00923779 27.7126745 4 Library prep * Sequenc-
ing depth -0.0810149 0.01255696 -6.4517919 p OR CI 25% CI 75% 1 6.267115e-01 2.522286 0.06061644
104.9538493 2 1.113863e-01 7.185244 0.63390479 81.4439797 3 4.912278e-169 1.291758 1.26857955 1.3153594
4 1.105353e-10 0.922180 0.89976061 0.9451579

```

```

$Random_Effects Random_Effect Variance SD 1 Virus target 22.106265 4.70173 2 Sample 7.604634 2.75765

```

```

# Optional: use kable() to print nicely in RMarkdown
kable(fixed_tbl, digits = 3, caption = "Fixed effects from GLMM (binomial)")

```

Fixed effects from GLMM (binomial)

Effect	Estimate	SE	z	p	OR	CI 25%	CI 75%
intercept	0.925	1.902	0.486	0.627	2.522	0.061	104.954
Library prep	1.972	1.239	1.592	0.111	7.185	0.634	81.444
Sequencing depth	0.256	0.009	27.713	0.000	1.292	1.269	1.315
Library prep * Sequencing depth	-0.081	0.013	-6.452	0.000	0.922	0.900	0.945

```

kable(random_tbl, digits = 3, caption = "Random effects (variance components) from GLMM")

```

Random effects (variance components) from GLMM

Random_Effect	Variance	SD
Virus target	22.106	4.702
Sample	7.605	2.758

Viral sequencing depth plots

```

# Create a grid of sequencing depths for prediction
depth_grid <- seq(min(det_long$depth_scaled),
                  max(det_long$depth_scaled),
                  length.out = 100)

newdata <- expand.grid(
  library_prep = c("MSSPE", "NGS"),
  depth_scaled = depth_grid
)

# Predicted probabilities and 95% CI using emmeans
emm <- emmeans(model_seq_depth, ~ library_prep | depth_scaled,
               at = list(depth_scaled = depth_grid), type="response")

```



```

emm_df <- as.data.frame(emm)

# Rename columns for clarity
emm_df <- emm_df %>%
  rename(Detection_Prob = prob,
         CI_Lower = asymp.LCL,
         CI_Upper = asymp.UCL)

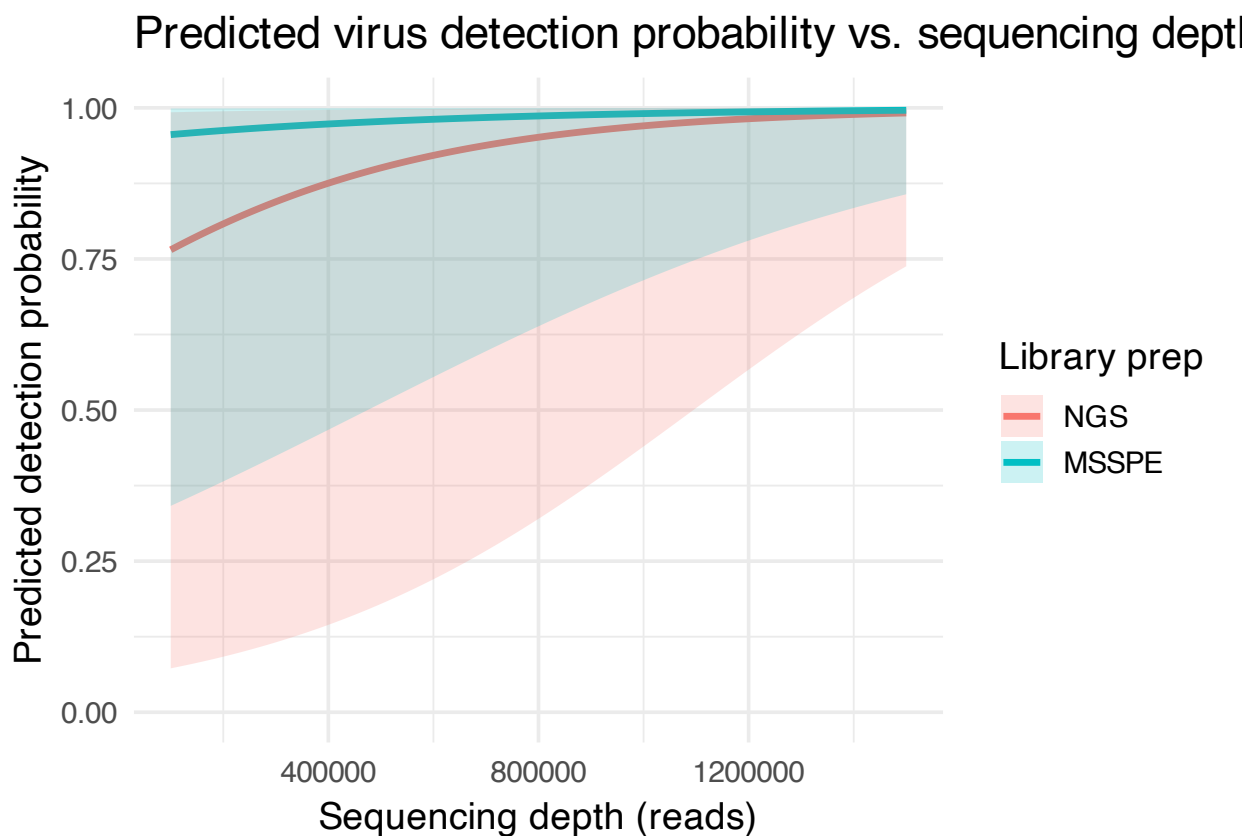
# Plot
ggplot(emm_df, aes(x = depth_scaled*1e5, y = Detection_Prob, color = library_prep)) +
  geom_line(size = 1.2) +
  geom_ribbon(aes(ymin = CI_Lower, ymax = CI_Upper, fill = library_prep), alpha = 0.2, color = NA) +
  scale_x_continuous(name = "Sequencing depth (reads)") +
  scale_y_continuous(name = "Predicted detection probability", limits = c(0,1)) +
  labs(color = "Library prep", fill = "Library prep") +
  theme_minimal(base_size = 14) +
  ggtitle("Predicted virus detection probability vs. sequencing depth")

```

```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```



Tables

```
cat("\\captionsetup{labelformat=empty}")
```

```
det_summary_libprep_box_format <- det_summary_libprep %>%
  group_by(depth, library_prep, target) %>%
  summarise(mean_prob = mean(detect_prob, na.rm = TRUE),
            .groups = "drop")

det_summary_libprep_box_format
```

A tibble: 240 x 4

```
depth library_prep target mean_prob
<dbl> <chr>          <chr>      <dbl>
```

```
1 100000 MSSPE EVG 0.43 2 100000 MSSPE PAsV 1
3 100000 MSSPE PBoV 1
4 100000 MSSPE PCV 0.53 5 100000 MSSPE PKV 0
6 100000 MSSPE PSapeV 0.34 7 100000 MSSPE PSapoV 0.69 8 100000 MSSPE PTeV 0.5 9 100000 NGS
EVG 0.29 10 100000 NGS PAsV 1
# i 230 more rows
```

```
det_summary_wide <- det_summary_libprep_box_format %>%
  group_by(target, depth, library_prep) %>% # average across library preps if multiple
  summarise(mean_prob = mean(mean_prob), .groups = "drop") %>%
  mutate(
    depth = paste0(depth / 1e6, "M")
  ) %>%
  pivot_wider(
    names_from = c(depth),
    values_from = mean_prob
  )

det_summary_wide
```

A tibble: 16 x 17

```
target library_prep 0.1M 0.2M 0.3M 0.4M 0.5M 0.6M 0.7M 0.8M 1 EVG MSSPE 0.43 0.7 0.75 0.83 0.83 0.89
0.9 0.9 2 EVG NGS 0.29 0.54 0.65 0.65 0.68 0.73 0.77 0.84 3 PAsV MSSPE 1 1 1 1 1 1 1
4 PAsV NGS 1 1 1 1 1 1 1 1
5 PBoV MSSPE 1 1 1 1 1 1 1 1
6 PBoV NGS 0.86 0.92 0.95 0.99 0.99 1 1 1
7 PCV MSSPE 0.53 0.72 0.79 0.9 0.9 0.9 0.9 0.9 8 PCV NGS 0.36 0.54 0.66 0.66 0.72 0.72 0.76 0.74 9 PKV
MSSPE 0 0.02 0.08 0.1 0.16 0.25 0.28 0.27 10 PKV NGS 0 0 0 0 0 0.01 0.01 11 PSapeV MSSPE 0.34 0.53
0.6 0.6 0.6 0.59 0.6 0.6 12 PSapeV NGS 0.17 0.46 0.61 0.65 0.7 0.74 0.74 0.73 13 PSapoV MSSPE 0.69 0.82
0.87 0.88 0.9 0.9 0.9 0.9 14 PSapoV NGS 0.51 0.71 0.9 0.94 0.98 0.98 1 1
15 PTeV MSSPE 0.5 0.57 0.66 0.7 0.7 0.7 0.7 0.7 16 PTeV NGS 0.36 0.47 0.57 0.62 0.61 0.64 0.67 0.7 # i 7
more variables: 0.9M , 1M , 1.1M , 1.2M , # 1.3M , 1.4M , 1.5M
```

Detection probability per viral target at varying sequencing depths

Virus	Library Prep	0.1M	0.3M	0.5M	0.7M	0.9M	1.1M	1.3M	1.5M
EVG	MSSPE	0.43	0.75	0.83	0.90	0.90	0.90	0.90	0.90
EVG	NGS	0.29	0.65	0.68	0.77	0.84	0.89	0.89	0.90
PAstV	MSSPE	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PAstV	NGS	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PBoV	MSSPE	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PBoV	NGS	0.86	0.95	0.99	1.00	1.00	1.00	1.00	1.00
PCV	MSSPE	0.53	0.79	0.90	0.90	0.90	0.90	0.90	0.90
PCV	NGS	0.36	0.66	0.72	0.76	0.77	0.80	0.79	0.79
PKV	MSSPE	0.00	0.08	0.16	0.28	0.30	0.30	0.30	0.30
PKV	NGS	0.00	0.00	0.00	0.01	0.03	0.03	0.04	0.10
PSapeV	MSSPE	0.34	0.60	0.60	0.60	0.60	0.60	0.60	0.60
PSapeV	NGS	0.17	0.61	0.70	0.74	0.76	0.77	0.78	0.80
PSapoV	MSSPE	0.69	0.87	0.90	0.90	0.90	0.90	0.90	0.90
PSapoV	NGS	0.51	0.90	0.98	1.00	0.99	1.00	1.00	1.00
PTeV	MSSPE	0.50	0.66	0.70	0.70	0.70	0.70	0.70	0.70
PTeV	NGS	0.36	0.57	0.61	0.67	0.69	0.70	0.70	0.70

```

det_summary_wide %>%
  rename(
    "Virus" = target,
    "Library Prep" = library_prep
  ) %>%
  select(-`0.2M`, -`0.4M`, -`0.6M`, -`0.8M`, -`1M`, -`1.2M`, -`1.4M`) %>%
  kable(
    format = "latex", # or "latex" if using LaTeX
    caption = "Detection probability per viral target at varying sequencing depths",
    digits = 2,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )

```

```

# Summarized detection probabilities from simulation
res_summary_2 <- summarize_detection(res_targets, detection_threshold = 10)

# Access outputs
det_long_2 <- res_summary$long # replicate-level total reads + detected 0/1
det_summary_2 <- res_summary$summary # mean reads + detection prob per depth x target
det_summary_libprep_2 <- res_summary$summary_libprep # mean reads + detection prob per depth x target

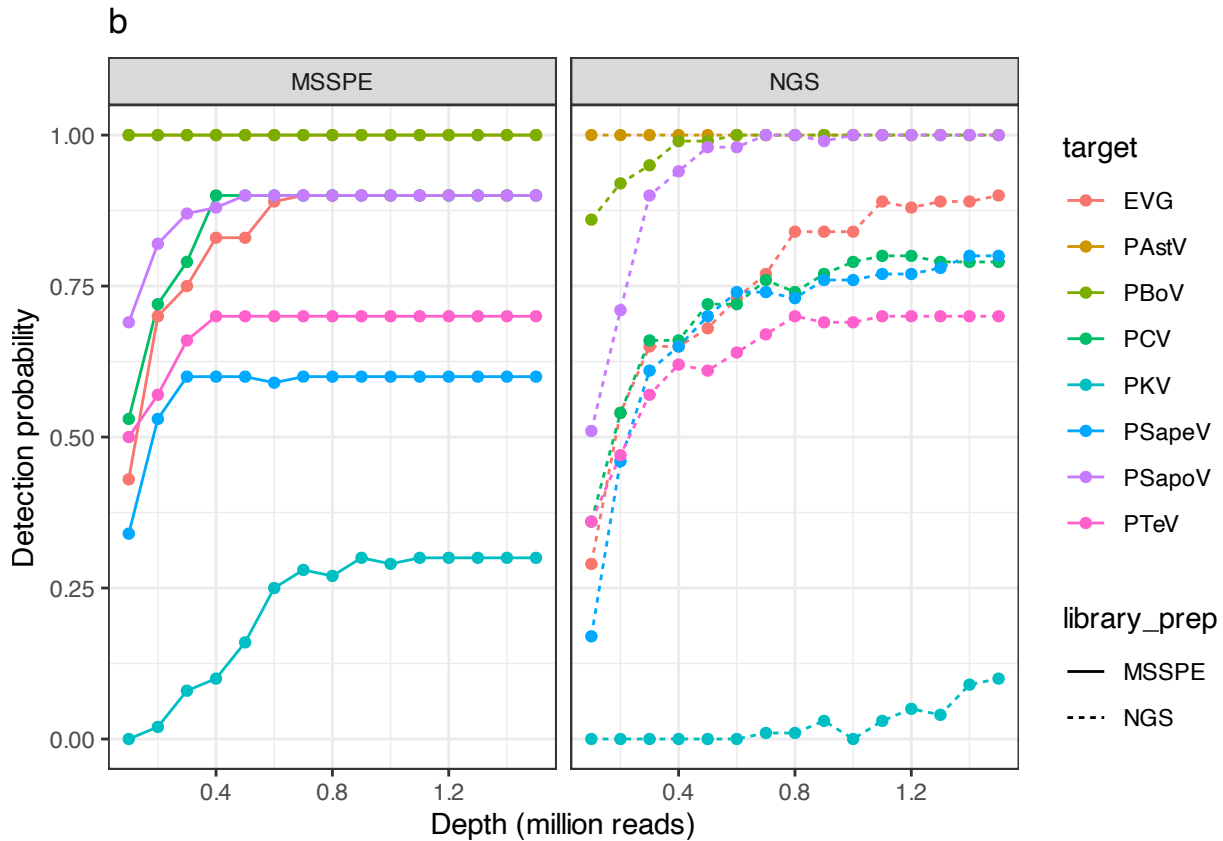
det_summary_libprep_box_format_2 <- det_summary_libprep_2 %>%
  group_by(depth, library_prep, target) %>%
  summarise(mean_prob = mean(detect_prob, na.rm = TRUE),
    .groups = "drop")

```

```

# Plot Library Prep Curve
rare_targets_curve_prep <- ggplot(det_summary_libprep_box_format, aes(x = depth/1e6, y = mean_prob, col = target)) +
  geom_line() + # small horizontal dodge to separate lines
  #geom_point(position = position_jitter(width = 0.3, height = 0.0)) + # jitter x slightly
  geom_point() +
  facet_wrap(~library_prep) +
  labs(x = "Depth (million reads)", y = "Detection probability",
       title = "b") +
  theme_bw()
rare_targets_curve_prep

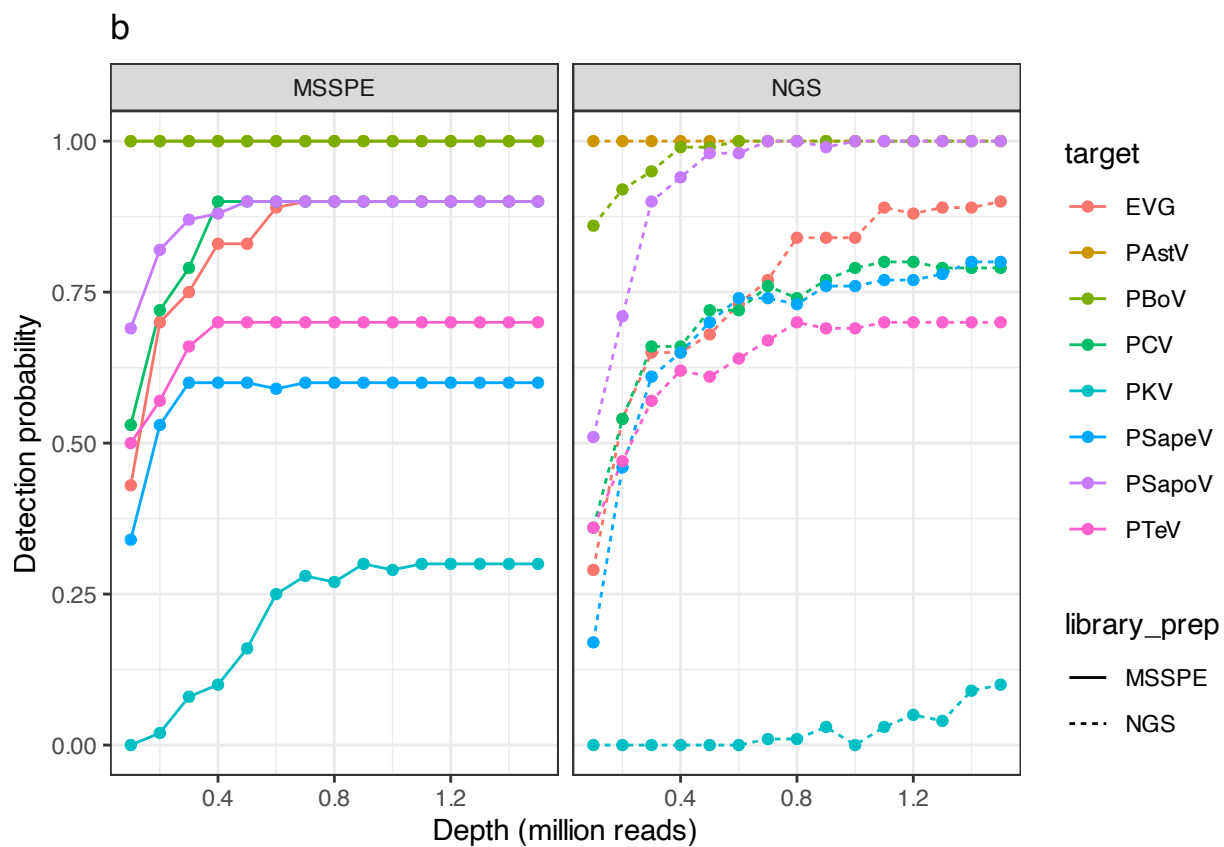
```



```

# Plot Library Prep Curve 2nd Depth
rare_targets_curve_prep_2 <- ggplot(det_summary_libprep_box_format_2, aes(x = depth/1e6, y = mean_prob,
  geom_line() + # small horizontal dodge to separate lines
  #geom_point(position = position_jitter(width = 0.3, height = 0.0)) + # jitter x slightly
  geom_point() +
  facet_wrap(~library_prep) +
  labs(x = "Depth (million reads)", y = "Detection probability",
       title = "b") +
  theme_bw()
rare_targets_curve_prep_2

```



```
#combined_target_curves = rare_targets_curve / rare_targets_curve_prep
#combined_target_curves
```